Appendix

On the replicability of LLM-based linguistic annotation: Insights from close and extended replications of Morin & Marttinen Larsson (2025) using open-weight LLMs

Full results

Distribution of classifications in the testing set provided by Morin & Marttinen Larsson (2025)

Table 1: Distribution of classifications in the testing set compared to the human gold standard

Classification	n	Percentage
evaluative	39	38%
non-evaluative	63	62%
Total	102	100%

Close replication

The following tables represent the results of the close replication.

Accuracy of all models compared to the human gold standard

Table 2: Classification accuracy and agreement with the human gold standard

Model	Accuracy	Alignments
Claude	93%	95/102
GPT	94%	96/102
Mistral	67%	68/102

Accuracy as 2x2 matrices

Table 3: GPT classifications compared to the human gold standard

Table 3: Prediction (GPT) vs Reference (Human)

	evaluative	non-evaluative
evaluative	37	4
non-evaluative	2	59

Table 4: Mistral classifications compared to the human gold standard

Table 4: Prediction (Mistral) vs Reference (Human)

	evaluative	non-evaluative
evaluative	7	2
non-evaluative	32	61

Table 5: Claude classifications compared with the human gold standard

Table 5: Prediction (Claude) vs Reference (Human)

	evaluative	non-evaluative
evaluative	37	5
non-evaluative	2	58

Accuracy metrics for all models in close replication

Table 6: Classification performance of the three models (Mistral, GPT and Claude) compared to the human gold standard

Metric	Mistral	GPT	Claude (original study)
Accuracy	67%	94%	93%
Precision	77.8%	90.2%	88.1%
Recall	17.9%	94.9%	94.9%
F1-score	29.2%	92.5%	91.4%
Matthew's correlation coefficient	0.25	0.88	0.86

Extended replication with GPT: reduced training process

Accuracy compared to the human gold standard

Table 7: GPT performance in extended replication (reduced training: only pretraining dataset and testing)

Accuracy	Alignments
94%	96/102

Accuracy as 2x2 matrices

Table 8: GPT classifications with few-shot training compared to the human gold standard

Table 8: Prediction (GPT reduced training) vs Reference (Human)

	evaluative	non-evaluative
evaluative	39	6
non-evaluative	0	57

Accuracy metrics of GPT with reduced training compared to Claude in original study

Table 9: Classification performance of GPT after reduced training and Claude compared to the human gold standard

Metric	GPT (reduced training)	Claude (original study)
Accuracy	94%	93%
Precision	86.7%	88.1%
Recall	100%	94.9%
F1-score	92.9%	91.4%
Matthew's correlation coefficient	0.89	0.86

Further details on the study

Technical details regarding the models

The open LLMs were accessed remotely via an API (KI:connect). The generation parameters in the API call were set to *temperature* = 0, $top_p = 1$, and a constant random seed was used. Setting *temperature* to 0, meaning the model always chooses the most probable next token, removes random sampling and therefore makes additional constraints such as top_k (sampling only from the k most probable tokens) unnecessary and potentially confounding. Even though these measures were taken to improve reproducibility, identical outputs across runs cannot be guaranteed.

Data leakage

For this study, it is assumed that the models, functioning as several "annotators", independently label the data without knowledge of each other's work. It can only be *assumed* because there is a possibility that the models might have the gold standard provided in the original study as part of their training data, which would mean they would not be totally independent. The Mistral model used for this study was released in June 2025 (MistralAI 2026), and the GPT model in August 2025 (OpenAI 2026). Morin and Marttinen Larsson (2025), who used data from the NOW corpus and Sketch Engine's EnTenTen21 corpus for their study (Morin and Marttinen Larsson 2025: 3), created their OSF-database containing the (annotated) datasets in November 2024 (Morin and Marttinen Larsson 2024). When and on which data the models were trained exactly remains unclear, but since they were released after the repository went public, the possibility of data leakage cannot be entirely ruled out.

References

KI:connect. "KI:connect API Reference." https://chat.kiconnect.nrw/app/api-docs/.

MistralAI. 2026. "Mistral Small 3.2 24B Instruct 2506 [Large Language Model]." https://docs.mistral.ai/models/mistral-small-3-2-25-06.

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OpenAI. 2026. "GPT OSS 120B [large language model]." https://openai.com/de-DE/index/introducing-gpt-oss/.